

WASHINGTON WARREN FIELD, NC, USA

WMO: 746925

Lat: 35.570N

Lon: 77.050W

Elev: 12

StdP: 101.19

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 03741

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.2	-3.8	-23.4	0.5	6.6	-18.0	0.8	4.1	9.3	14.7	8.2	12.0	1.7	350	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	34.7	25.7	33.7	25.0	32.3	24.7	27.7	31.5	26.8	30.7	26.0	30.0	3.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.7	30.1	26.0	21.3	29.1	24.9	19.9	28.4	87.9	31.8	83.5	30.1	80.6	30.2	32.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	7.0	5.7	-9.2	36.7	1.8	1.2	-10.5	37.6	-11.6	38.3	-12.6	39.0	-14.0	39.9		
(5)			-11.1	28.9	2.7	1.6	-13.1	30.1	-14.7	31.0	-16.2	31.9	-18.2	33.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	6.3	7.8	11.4	16.8	21.2	25.3	27.1	26.1	23.1	17.7	12.0	8.7
	DBStd	8.50	6.16	5.32	5.44	4.62	3.92	2.88	2.40	2.52	2.93	4.50	4.90	5.35
	HDD10.0	425	149	97	49	4	0	0	0	0	0	3	33	91
	HDD18.3	1566	374	298	222	83	19	1	0	0	3	67	197	303
	CDD10.0	2978	34	34	94	207	346	460	529	498	392	241	93	49
	CDD18.3	1079	2	3	8	37	107	211	271	240	145	47	7	3
	CDH23.3	9776	3	15	72	327	907	1976	2779	2249	1069	329	39	10
	CDH26.7	3476	0	1	7	69	291	760	1138	846	297	65	2	0
Wind	WSAvg	2.3	2.4	2.7	2.9	2.9	2.6	2.2	2.2	1.8	2.2	2.0	2.0	2.1
Precipitation	PrecAvg	1243	93	75	108	79	101	109	133	131	159	93	84	82
	PrecMax	1665	171	167	174	146	206	279	242	292	665	309	218	202
	PrecMin	915	32	29	37	33	44	7	60	67	24	1	17	24
	PrecStd	201	38	38	36	34	43	61	40	56	146	63	50	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	26.0	27.6	30.6	34.2	36.2	37.2	35.2	33.0	31.0	27.1	25.1
		MCWB	19.0	19.4	17.4	19.6	24.0	24.2	25.3	26.9	24.6	23.9	19.6	20.7
	2%	DB	21.8	22.4	25.5	28.3	31.9	33.9	34.8	33.9	31.1	28.2	24.1	22.4
		MCWB	17.8	16.5	17.4	19.0	23.1	24.5	26.4	25.7	23.6	21.5	18.1	18.4
	5%	DB	18.0	19.5	23.0	27.2	29.8	32.7	33.7	32.5	29.6	27.1	22.4	20.0
		MCWB	14.6	15.3	16.2	18.8	22.1	24.5	25.5	25.1	22.8	20.8	17.9	17.4
	10%	DB	16.3	17.4	20.9	25.4	27.8	31.0	32.2	31.0	28.0	25.2	20.3	17.7
		MCWB	12.9	12.6	15.7	18.5	21.6	24.3	25.0	24.7	22.5	19.8	16.8	14.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.5	19.8	21.3	24.7	27.6	28.9	29.3	28.7	26.7	25.1	22.2	21.4
		MCDB	23.1	24.1	24.7	26.9	31.3	31.8	34.8	34.6	29.0	28.4	25.2	23.8
	2%	WB	18.0	18.2	19.4	22.7	25.4	27.3	27.7	27.1	25.4	23.8	20.2	19.4
		MCDB	20.7	21.4	23.5	25.4	28.4	29.6	31.9	31.9	28.8	27.1	22.5	21.4
	5%	WB	15.7	16.3	18.0	21.1	23.9	26.2	26.8	26.1	24.6	22.6	18.9	17.4
		MCDB	18.1	19.1	21.3	24.6	27.1	29.0	31.4	30.7	28.1	25.7	21.7	19.3
	10%	WB	13.6	13.6	16.5	19.8	22.8	25.4	25.9	25.4	23.8	21.1	17.0	15.2
		MCDB	15.8	16.3	19.9	23.9	26.5	29.2	30.4	29.6	27.3	24.0	19.7	17.6
Mean Daily Temperature Range	5% DB	MDBR	10.6	11.2	11.7	11.9	10.7	9.8	9.2	9.1	9.3	10.9	11.8	10.7
		MCDBR	13.2	14.1	14.1	13.4	12.3	11.2	10.3	10.0	10.6	11.5	12.8	11.8
		MCWBR	10.2	9.7	8.3	6.8	5.5	5.0	4.8	5.0	5.1	6.7	8.5	9.1
	5% WB	MCDBR	12.1	13.2	11.7	11.3	10.5	9.5	9.2	8.9	8.3	8.9	10.7	10.8
		MCWBR	10.5	10.6	8.5	7.7	6.0	5.4	4.9	5.0	4.8	6.1	8.5	9.2
Clear-Sky Solar Irradiance	taub	0.310	0.326	0.353	0.395	0.443	0.497	0.526	0.515	0.416	0.362	0.329	0.316	
	taud	2.531	2.480	2.432	2.320	2.222	2.115	2.048	2.070	2.348	2.480	2.526	2.546	
	Ebn at Noon	896	917	917	890	847	798	772	772	840	864	864	865	
	Edh at Noon	80	94	108	127	142	158	168	161	116	93	80	75	
All-Sky Solar Radiation	RadAvg	2.57	3.28	4.31	5.46	6.03	6.34	6.04	5.47	4.53	3.72	2.84	2.28	
	RadStd	0.17	0.35	0.29	0.37	0.44	0.34	0.33	0.31	0.32	0.45	0.21	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+156	+110

Nomenclature: See separate page